

**MATHEMATICS 2**

**40 MINUTES**

**INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## **Mathematics 2**

- 1 What is the value of the definite integral

$$\int_{-2}^2 (3x^2 + 5 + x^3 \cos x) dx$$

- A 0
- B 16
- C 18
- D 36
- E 72

- 2 The diagram shows a rectangle of length  $L$ , width  $W$ , and area  $A$ .

Given that

$$A = \frac{3x^2 - 11x - 4}{9x^2 - 1} \quad \text{and} \quad W = \frac{x^2 - 2x - 8}{3x^2 + 14x - 5}$$

which one of the following is an expression for  $L$  in its simplest form?

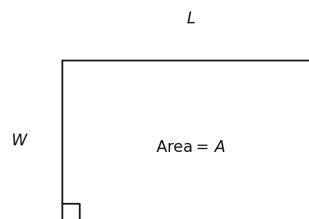


Diagram not to scale

- A  $\frac{x-5}{x+2}$
- B  $\frac{x-5}{x-2}$
- C  $\frac{x+2}{x+5}$
- D  $\frac{x+5}{x-2}$
- E  $\frac{x+5}{x+2}$

- 3 The function  $f(x)$  is continuous on the interval  $0 \leq x \leq 8$ . The curve  $y = f(x)$  crosses the  $x$ -axis exactly once in this interval.

Given that:

$$\int_0^8 f(x) \, dx = 2$$

and the total area of the region bounded by the curve  $y = f(x)$ , the  $x$ -axis, and the lines  $x = 0$  and  $x = 8$  is 12.

What is the area of the region lying below the  $x$ -axis?

- A 2  
B 5  
C 6  
D 7  
E 10
- 4 The table shows the number of items and the mean score for three groups of data, A, B, and C.

| Group | Number of items | Mean score |
|-------|-----------------|------------|
| A     | $n$             | 12         |
| B     | $2n$            | 18         |
| C     | $k$             | 30         |

When all three groups are combined, the overall mean score of all the items is 20.

What is the ratio  $k : n$ ?

- 5 A curve has equation  $y = f(x)$ . The derivative of the function is given by  $f'(x) = 3x^2 - 12$ . The distance between the two stationary points of the curve is  $4\sqrt{65}$ .

What is the  $y$ -intercept of the curve?

- A -16  
B 0  
C 16  
D 32  
E Cannot be determined

- 6 The diagram shows two perpendicular straight lines,  $L_1$  and  $L_2$ , in the Cartesian plane.

Line  $L_1$  passes through the points with coordinates  $(2 - k, 3k - 1)$  and  $(k + 4, k + 5)$ , where  $k$  is a constant.

Line  $L_2$  intersects the coordinate axes at  $(0, 4)$  and  $(6, 0)$ .

What is the value of  $k$ ?

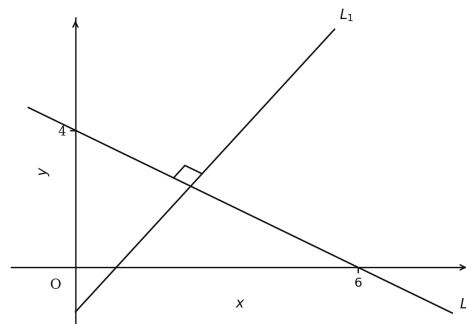


Diagram not to scale

- A**  $-9$
- B**  $-\frac{3}{5}$
- C**  $\frac{3}{5}$
- D**  $\frac{7}{5}$
- E**  $\frac{5}{3}$
- F**  $11$
- 7 What is the sum of all real values of  $x$  that satisfy the equation  $\sin(2x) = \cos(x)$  in the interval  $0 \leq x \leq \pi$ ?
- A**  $\frac{\pi}{2}$
- B**  $\frac{2\pi}{3}$
- C**  $\pi$
- D**  $\frac{4\pi}{3}$
- E**  $\frac{3\pi}{2}$

- 8 A rhombus has diagonals of length  $2\sqrt{5}$  cm and 4 cm. An enlargement of the rhombus has an area of  $80\sqrt{5}$  cm<sup>2</sup>.

What is the perimeter of the enlarged rhombus?

- A  $6\sqrt{5}$  cm
- B  $12\sqrt{5}$  cm
- C  $24\sqrt{5}$  cm
- D  $48\sqrt{5}$  cm
- E 240 cm
- F 480 cm

- 9 Given that  $\sin(\theta) = \frac{3}{5}$  and  $\frac{\pi}{2} < \theta < \pi$ , find the exact value of  $\cot(2\theta)$ .

- A  $\frac{24}{7}$
- B  $\frac{7}{24}$
- C  $-\frac{7}{24}$
- D  $-\frac{24}{7}$
- E  $-\frac{7}{12}$
- F  $-\frac{32}{21}$

- 10 The positive variables  $u$ ,  $v$ ,  $w$ , and  $x$  satisfy the following relationships:

- $u$  is inversely proportional to the square root of  $v$
- $v$  is directly proportional to the cube of  $w$
- $w$  is inversely proportional to the square of  $x$

Which of the following correctly describes the relationship between  $u$  and  $x$ ?

- A  $u$  is directly proportional to  $x^3$
- B  $u$  is inversely proportional to  $x^3$
- C  $u$  is directly proportional to  $x^6$
- D  $u$  is inversely proportional to  $x^6$
- E  $u$  is directly proportional to  $x^{3/2}$
- F  $u$  is inversely proportional to  $x^{3/2}$

- 11** A geometric series has a common ratio  $r$ . The value of  $r$  is the result of increasing the number 1 by  $p\%$ , and then decreasing the new value by  $p\%$ , where  $p > 0$ .  
The sum to infinity of this series exists.  
Which of the following gives the complete set of possible values for  $p$ ?
- A**  $0 < p < 100$
  - B**  $0 < p < 100\sqrt{2}$
  - C**  $p > 0$
  - D**  $p > 100\sqrt{2}$
  - E**  $0 < p < 200$
  - F** No values of  $p$ .
- 12** The expression  $(3x + 2)^6$  is expanded in ascending powers of  $x$ .  
How many of the coefficients in this expansion are divisible by 6?
- A** 4
  - B** 5
  - C** 6
  - D** 7
  - E** 2
- 13** The function  $f(x) = ax^2 + bx + c$  has a stationary point at  $x = 2$ . The gradient of the curve  $y = f(x)$  at  $x = 0$  is 12. Given that  $f(1) = 15$ , what is the value of  $c$ ?
- A** 0
  - B** 6
  - C** 9
  - D** 12
  - E** 24

- 14 A circle has the equation

$$x^2 + y^2 - 2x + 2y - 23 = 0$$

Find the area of the triangle bounded by the coordinate axes and the tangent to the circle at the point  $(4, 3)$ .

- A 6
- B 24
- C 27
- D 48
- E  $\frac{49}{24}$

- 15 The function  $f(x)$  is defined for  $0 \leq x \leq 6$  by:

$$f(x) = \begin{cases} 11 + 2x - x^2 & \text{for } 0 \leq x \leq 3 \\ 8x - x^2 - 6 & \text{for } 3 < x \leq 6 \end{cases}$$

What is the maximum value of  $f(x)$ ?

- A 9
- B 10
- C 11
- D 12

- 16 A function  $f$  is defined by

$$f(x) = \log_3 \left( \frac{ax + b}{x} \right)$$

where  $a$  and  $b$  are constants.

Given that  $f(1) = 2$  and  $f(4) = 1$ , find the value of  $x$  such that  $f(x) = -1$ .

- A -12
- B -6
- C -4
- D -2
- E 4

- 17** How many solutions does the equation  $2 \cos \left( 2x + \frac{\pi}{3} \right) = 1$  have in the interval  $0 \leq x \leq 2\pi$ ?
- A** 2  
**B** 3  
**C** 4  
**D** 5  
**E** 6
- 18** Quantities  $X$  and  $Y$  change in discrete steps  $n = 0, 1, 2, \dots$ . Initially,  $X_0 = 20Y_0$ , where  $Y_0 > 0$ . In each step,  $X$  decreases by 60% of its value at the start of the step, and  $Y$  decreases by 20% of its value at the start of the step. What is the minimum value of  $n$  such that  $X_n < Y_n$ ?
- A** 2  
**B** 4  
**C** 5  
**D** 6
- 19** The curve with equation  $y = 4^x - 10 \cdot 2^x$  intersects the horizontal line  $y = c$  at two distinct points. The  $x$ -coordinates of these points are  $p$  and  $q$ , where  $q > p$ . Given that  $q - p = 2$ , what is the value of the constant  $c$ ?
- A** -25  
**B** -24  
**C** -16  
**D** -4  
**E** 16
- 20** A circle passes through the points  $(2, 1)$  and  $(4, 5)$  and is tangent to the  $x$ -axis. What is the largest possible radius of the circle?
- A** 1  
**B** 2.5  
**C** 3  
**D** 5  
**E**  $5\sqrt{2}$

- 21** Let  $n$  be a positive odd integer. Two distinct coefficients in the expansion of  $(1 + x)^n$  are chosen at random.

The probability that these two coefficients are equal is  $\frac{1}{23}$ .

What is the value of  $n$ ?

- A** 21
- B** 22
- C** 23
- D** 24
- E** 25
- F** 45

- 22** A sequence of three positive integers has a sum of 6 and a sum of the squares of its terms of 18.

How many such distinct sequences are there?

- A** 0
- B** 1
- C** 2
- D** 3
- E** 6
- F** 9

- 23** A convergent geometric series has first term  $\tan \theta$  and third term  $\cot \theta$ . The second term of the series is negative, and the sum to infinity is 1.

What is the value of the first term?

- A**  $\frac{1}{2}$
- B** 1
- C**  $\frac{1 + \sqrt{5}}{2}$
- D**  $\frac{1 - \sqrt{5}}{2}$
- E**  $\frac{-1 + \sqrt{5}}{2}$

- 24** A particle travels at a constant speed  $v$  from the point  $(-2R, 0)$  to the point  $(2R, 0)$ , where  $R > 0$ .

The particle's path is chosen to be the shortest possible route that does not enter the region  $x^2 + y^2 < R^2$ .

What is the minimum time for this journey?

- A**  $\frac{R}{v} (2\sqrt{3} + \frac{\pi}{3})$
- B**  $\frac{R}{v} (\pi + 2)$
- C**  $\frac{R}{v} (2\sqrt{3} + \frac{2\pi}{3})$
- D**  $\frac{R}{v} (2\sqrt{5})$
- E**  $\frac{R}{v} (2\sqrt{3})$

- 25** A function  $f$  is defined by

$$f(x) = \frac{x^2 + 5x + 4}{x^2 + 7x + 12} \div \frac{x + 1}{3x + 9}$$

for all values of  $x$  for which the expression is defined.

What is the value of  $f'(4)$ ?

- A** 0
- B**  $\frac{1}{3}$
- C** 1
- D** 3
- E** 4
- F** -3

- 26** A geometric progression has first term  $a$  and common ratio  $r$ .

The first term is

$$a = \frac{1}{x+2}$$

and the sum to infinity is

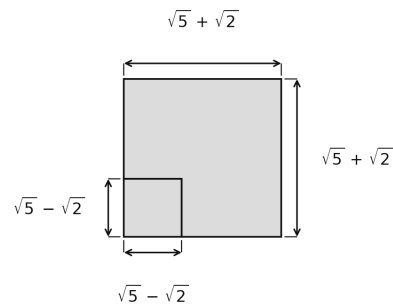
$$S_{\infty} = \frac{x-1}{x^2-4}$$

What is the common ratio  $r$ ?

- A**  $\frac{1}{x-1}$
- B**  $\frac{x-2}{x-1}$
- C**  $\frac{1}{2-x}$
- D**  $\frac{2x-3}{x-1}$
- E**  $-\frac{1}{x-1}$
- F**  $\frac{x+1}{x+2}$

- 27** The diagram shows a square of side length  $\sqrt{5} + \sqrt{2}$ . A smaller square of side length  $\sqrt{5} - \sqrt{2}$  is removed from one corner, leaving the shaded region.

What is the area of the shaded region?



*Diagram not to scale*

- A** 4
- B** 8
- C** 14
- D**  $2\sqrt{10}$
- E**  $4\sqrt{10}$
- F**  $7 + 2\sqrt{10}$
- G**  $14 - 4\sqrt{10}$
- H**  $14 + 4\sqrt{10}$